Bottle Sealing Machine



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Automatic bottle sealing unit; Soy milk bottle sealing machine manual

What it is used for

A bottle sealing machine closes bottles using a compatible cap, liner, or sealing film to help prevent leakage, contamination, and product spoilage. Correct setup, seal selection, hygienic handling, and seal-quality checks help maintain packaging integrity during storage and transport.

Main parts

- Machine frame supports the sealing assembly.
- Bottle holder or conveyor positions bottles for sealing.
- Capping head or sealing head applies the closure.
- Heating element or induction unit activates compatible sealing liners where fitted.
- Temperature and time controls regulate heat sealing where applicable.
- Pressure or height adjustment sets the sealing force or position.
- Control panel and switches operate the machine.
- Safety guards protect operators from moving or hot parts.
- Power supply and cables provide electrical connection.

Before you start

1. Only trained personnel should operate the bottle sealing machine.
2. Read the machine manual and confirm compatibility with the bottle, cap, liner, film, and product.
3. Inspect the frame, sealing head, guards, cables, controls, and bottle holder for damage.
4. Clean and sanitise food-contact parts according to the approved hygiene procedure.
5. Check that bottles and closures are clean, undamaged, and suitable for the intended product.
6. Verify the specified temperature, pressure, sealing time, and alignment settings for the packaging materials.
7. Wear appropriate PPE and keep the work area dry, clean, and free of obstructions.

How to use

1. Place the machine on a stable surface and connect power according to the manufacturer's instructions.
2. Install the correct sealing head, cap guide, or film holder for the packaging format.
3. Set the required temperature, sealing time, pressure, and height according to the validated packaging procedure.
4. Allow the heating system to reach its specified operating condition where applicable.
5. Fill and prepare bottles according to the product process, leaving the required headspace where specified.

6. Position the cap, liner, or sealing film correctly on the bottle opening.
7. Place the bottle in the holder or conveyor and align it with the sealing head.
8. Activate the sealing cycle using the designated control and keep hands clear of moving parts.
9. Remove the sealed bottle after the cycle finishes and the head has returned to its safe position.
10. Inspect the closure for alignment, wrinkles, incomplete bonding, leakage, or other defects.
11. Perform the prescribed seal-integrity test on representative samples and record process settings and results.
12. Stop the machine, isolate power, and allow hot components to cool before cleaning or adjustment.

⚠ Safety

- Keep hands and fingers away from the sealing head, moving parts, and hot surfaces.
- Never bypass guards, interlocks, or emergency stops.
- Do not seal containers that are cracked, unsuitable for the process, or contaminated around the sealing area.
- Use only approved packaging materials and settings compatible with the bottle and product.
- Disconnect power before clearing jams, changing components, or performing maintenance.
- Follow product-specific hygiene requirements and avoid contaminating bottle openings or closures.

Care & storage

- Clean and sanitise product-contact components after use according to the approved procedure.
- Remove adhesive, film, or product residue using manufacturer-approved cleaning methods.
- Inspect sealing heads, holders, guides, cables, and guards regularly.
- Check temperature and sealing controls at the prescribed maintenance intervals.
- Arrange authorised servicing if seals become inconsistent or the machine overheats.

Common mistakes to avoid

- Using an incompatible cap, liner, or sealing film.
- Applying incorrect temperature, pressure, or sealing time.
- Misaligning the bottle or closure during sealing.
- Skipping seal-integrity checks or cleaning between production runs.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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